

Lebanese University – Faculty of Engineering

Subject: Sensors

Professor: Dr. Mohammad Aoude



Sensors project
(Greenhouse)

Presented by:

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1. Summary of Contribution

I was responsible for the physical hardware layer of the greenhouse: wiring every sensor and actuator to the Arduino Mega 2560, planning the power distribution between the 5V and 3.3V rails, and building the signal-conditioning circuits (the LDR voltage divider, the DHT11 pull-up, and the 220Ω current-limiting resistors on all indicator LEDs). I also handled the electrical isolation of the 12V irrigation pump through the 2-channel relay module and the external 12V adapter, and I assembled the wooden and acrylic enclosure that houses the breadboard, LCD, and RFID reader shown in the final prototype photo. Toward the end of the project I helped trace and fix a systematic offset on the soil and water-level analog readings that turned out to be caused by resistance in long, unshielded jumper wires, which we corrected by shortening runs and re-routing the breadboard layout

2. Git Activity

- Add initial wiring diagram for sensor layout
- Wire DHT11 with 10k pull-up resistor
- Wire soil moisture and water level sensors to A1/A0
- Add relay module wiring for 12V irrigation pump
- Build enclosure mounts for LCD and RFID reader
- Fix analog offset caused by long jumper wire runs
- Add 220-ohm resistors to LED indicator circuits

3. Reflection

Working on the hardware side taught me how much small electrical details matter in a system this size — a single loose ground connection or an under-rated resistor could throw off readings across the whole board. Debugging the jumper-wire offset in particular was a good lesson in tracing a software-looking symptom back to a physical cause. I'd like to get more comfortable with PCB design so a future version of this project could replace the breadboard with something more robust.